

Amirmohammad (Amir) Mohammadi (U.S. Permanent Resident)

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EDUCATION

Texas A&M University, College Station, Texas	December 2026
Doctor of Philosophy in Computer Engineering	
Thesis: Distribution-guided Parameter-efficient Fine-tuning of Foundation Models for Waveform-based Sensing	
Sharif University of Technology, Tehran, Iran	February 2021
Master of Science in Electrical Engineering	
Thesis: Design and Evaluation of an Integrated Biomedical System for Human Stress Detection and Monitoring	
University of Tabriz, Tabriz, East Azerbaijan Province, Iran	September 2018
Bachelor of Science in Electrical Engineering	
Project: Development of a Data Logger for Sensors of Android Smartphones with Python	

PROFESSIONAL EXPERIENCE

Texas A&M University, College Station, Texas	September 2022 – December 2026
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Graduate Research Assistant

- **Built** a few-shot whole slide image classification pipeline to investigate parameter-efficient fine-tuning of vision-language models for disease diagnosis in digital pathology applications.
- **Developed** a distribution-aware adaptation algorithm for Transformer-based foundation models as a parameter-efficient fine-tuning technique for waveform-based sensing applications.
- **Improved** the classification accuracy of a convolutional-based deep learning model by designing a time-frequency feature combination and engineering pipeline for audio-based applications.
- **Reduced** ground truth data requirements by a factor of 15 for physiological time-series signals by developing physics-informed neural networks for cuffless blood pressure measurement.
- **Disseminated** research findings through publications, presentations, and reports to funding agencies.

Sharif University of Technology, Tehran, Iran	July 2019 – February 2021
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Graduate Student Researcher

- **Designed** an ECG + EDA wearable sensor with a 3× longer battery life compared to alternatives and delivered 94% mental stress detection accuracy across 18 participants using machine learning models.

SKILLS

- **Tools:** Python, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, MATLAB, C, Hugging Face, Transformers, Git, Linux, VS Code, Jupyter Notebook, High-performance Computing.
- **Interests:** Data Science, Applied AI, Foundation Models, Deep Learning Architectures & Techniques, Generative AI, Physics-Informed Machine Learning, Multi-Modal Learning, Efficient AI, AI Hardware.

PEER-REVIEWED PUBLICATIONS

- J. Ritu, **A. Mohammadi**, D. Carreiro, A. Van Dine, and J. Peebles, "Structural and Statistical Audio Texture Knowledge Distillation for Acoustic Classification," in *IEEE Access*, 2026. [[link](#)]

- **A. Mohammadi**, D. Carreiro, A. Van Dine, and J. Peeples, "Histogram-based Parameter-efficient Tuning for Passive and Active Sonar Classification," *IEEE International Geoscience and Remote Sensing Symposium*, 2026. [[link](#)]
- F. Orvati Nia, **A. Mohammadi**, S. Al Kharsa, P. Naikare, Z. Hampel-Arias, and J. Peeples, "Neighborhood Feature Pooling for Remote Sensing Image Classification," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*, March 2026, pp. 1393–1402. [[link](#)]
- **A. Mohammadi**, I. Masabarakiza, E. Barnes, D. Carreiro, A. Van Dine and J. Peeples, "Investigation of Time-Frequency Feature Combinations with Histogram Layer Time Delay Neural Networks," *OCEANS 2025 Brest*, 2025, pp. 1-6. [[link](#)]
- **A. Mohammadi**, T. Kelhe, D. Carreiro, A. Van Dine, and J. Peeples, "Cross-Domain Knowledge Transfer for Underwater Acoustic Classification Using Pre-trained Models," in *OCEANS 2025 Brest*, 2025, pp. 1-6. [[link](#)]
- K. Sel, **A. Mohammadi**, R. I. Pettigrew, and R. Jafari, "Physics-informed neural networks for modeling physiological time series for cuffless blood pressure estimation," *npj Digital Medicine*, vol. 6, no. 1, p. 110, June 2023. [[link](#)]
- **A. Mohammadi**, M. Fakharzadeh, and B. Baraeinejad, "An Integrated Human Stress Detection Sensor Using Supervised Algorithms," in *IEEE Sensors Journal*, vol. 22, no. 8, pp. 8216-8223, April 2022. [[link](#)]

MANUSCRIPTS UNDER REVIEW

- **A. Mohammadi**, A. Van Dine, and J. Peeples, "Parameter-efficient Dual-encoder Architecture with Differentiable Choquet Integral Fusion for Underwater Acoustic Classification". [[link](#)]
- **A. Mohammadi**, R. Khadka, K. A. Grant, M. L. Benton, and J. Peeples, "Predicting Ethanol Drinking Status from Nonhuman Primate Liver Histology: A Proof of Concept for Parameter-efficient Fine-tuning in Few-Shot Vision-Language Modeling".

LANGUAGES

English (proficient); **Farsi** (native/bilingual); **Azerbaijani Turkish** (native/bilingual)